

Breakup dynamics of Newtonian fluids under extension

Supplementary Materials

Ceri Allgood¹ (c.allgood1@lancaster.ac.uk)

Thomas J. Jones¹ (thomas.jones@lancaster.ac.uk)

¹ Lancaster Environment Centre, Lancaster University, Lancaster, UK

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SM1: Fluid properties

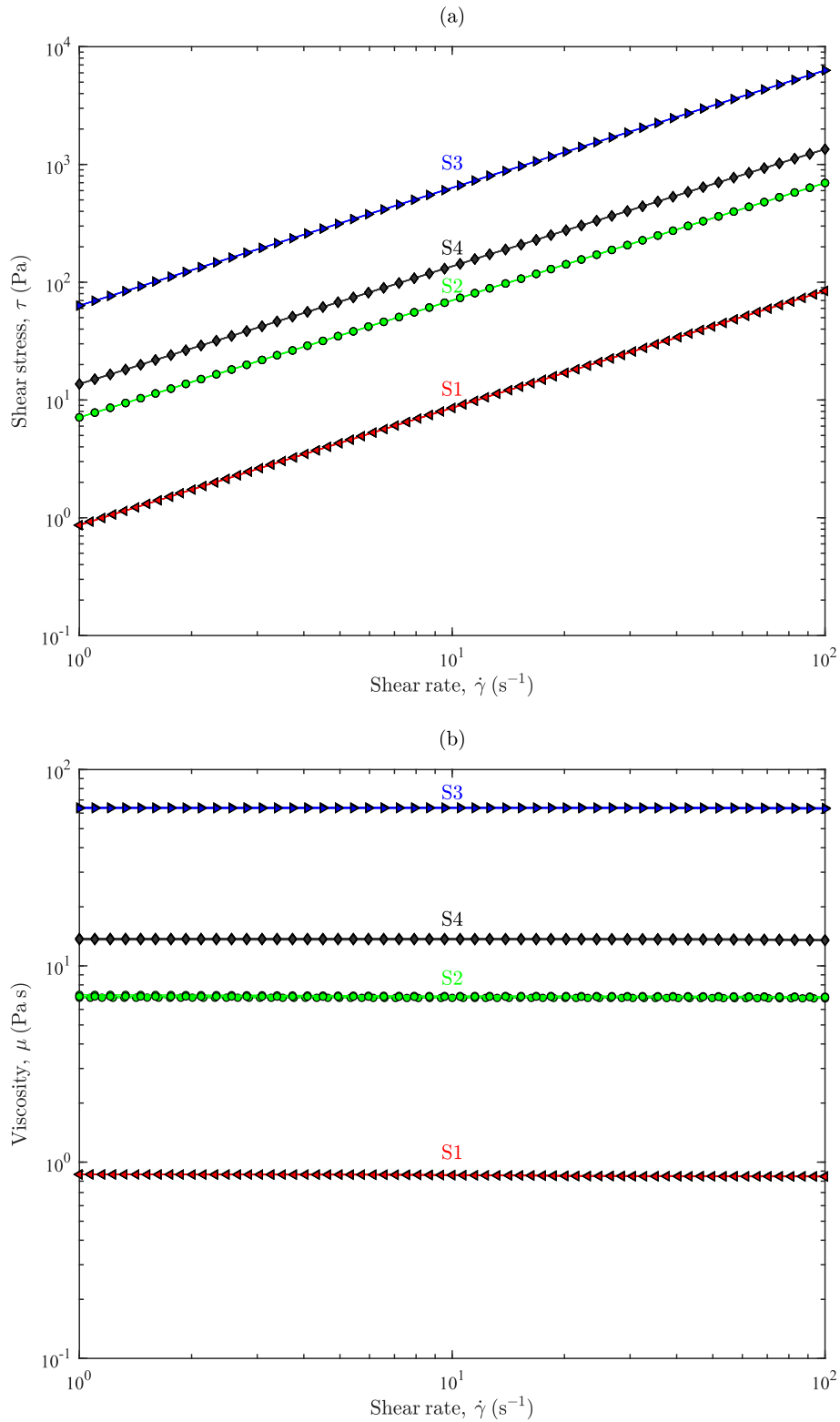


Figure A1. Shear rate sweeps on S1-4, on three separate aliquots of each fluid: **a)** shear rate vs. shear stress shows an approximately linear relationship, demonstrating Newtonian behaviour for all fluids; **b)** shear rate vs. viscosity. Fluid S4 (PDMS) displays minor shear-thinning as strain rates approach 100 s^{-1} . The other three fluids (polybutene) do not.

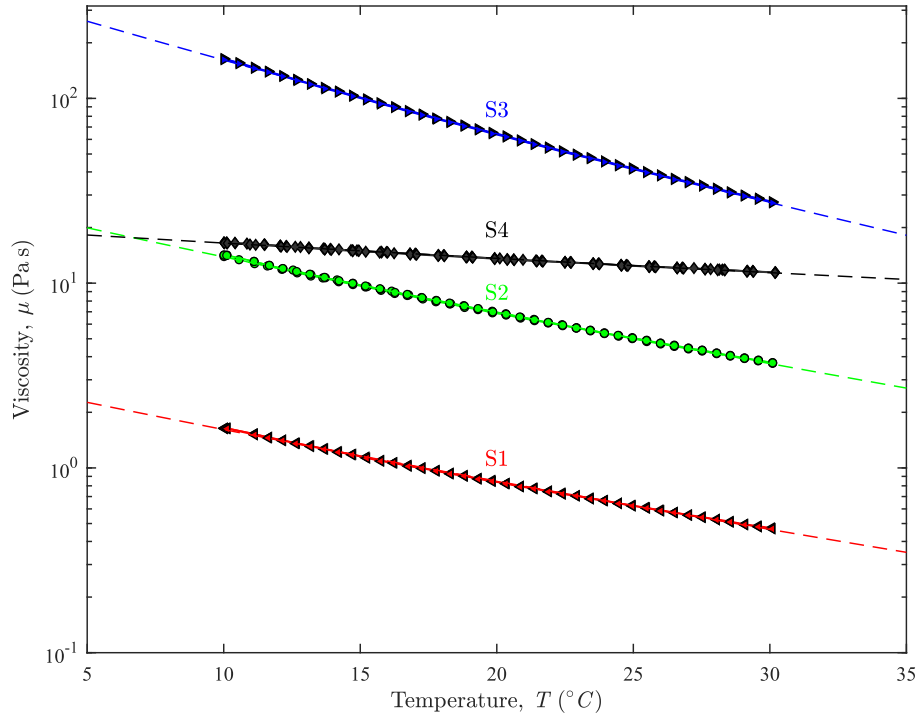


Figure A2. Viscosity-temperature relationships for four silicone oils S1-4, with dashed lines showing the best-fit equations reported in the main text (Eq. 14).

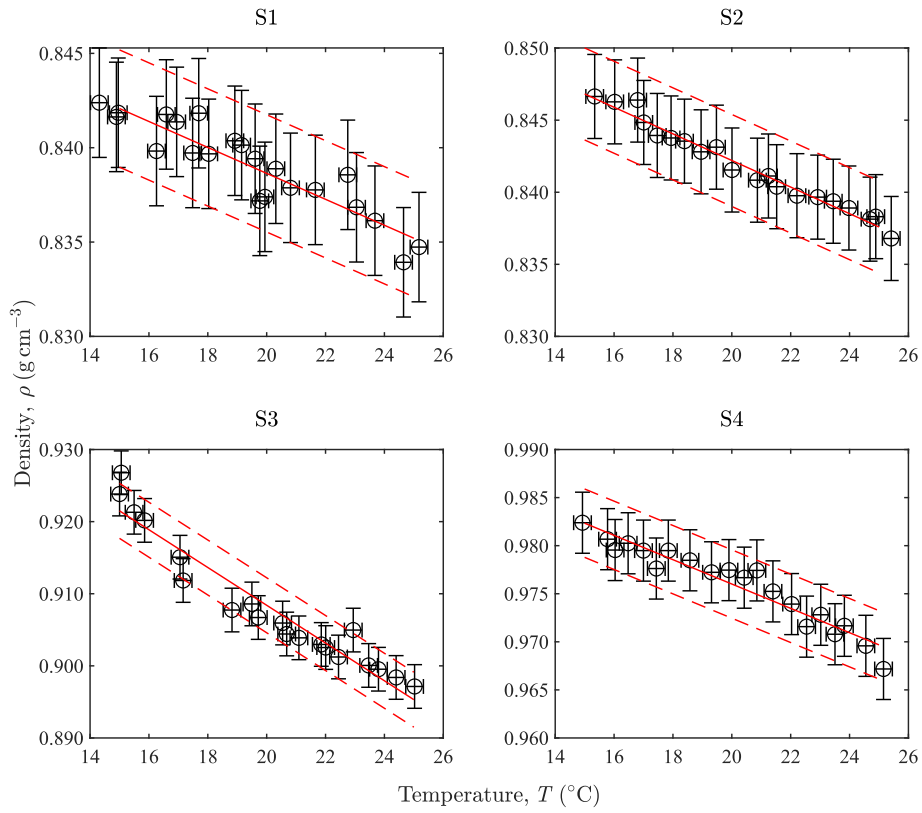


Figure A3. Density-temperature relationship for four silicone oils S1-4, with the solid red line showing the best-fit equations (Eq. 15) reported in Section 3.1 of the main text, and error bars and dashed red line showing the compound uncertainties associated with the measurement method, as outlined in Allgood and Jones (2025).

SM2: List of all experiments

Table A1. List of all experiments conducted, with details of the fluid type (S1-4); the diameter of the plates, which is equal to the initial diameter of the fluid column D_0 ; the temperature T of the experiment; the target velocity set on the TriMaster v_{aim} and the true velocity v_{calc} measured from the high-speed video footage; the initial fluid aspect ratio H_0/D_0 ; the breakup time t_{brk} observed from high-speed video footage; and the scale of the recorded image frames.

Name	Fluid	D_0 [mm]	T [°C]	v_{aim} [mm s ⁻¹]	v_{calc} [mm s ⁻¹]	H_0/D_0 [-]	t_{brk} [s]	Scale [px/ mm]
241113 m	S1	2	20.3	2	2.06	0.10	0.532	72.50
241113 n	S1	2	20.5	2	1.95	0.29	0.746	73.00
241113 o	S1	2	20.3	2	1.98	0.34	0.776	73.00
241113 p	S1	2	20.6	2	2.00	0.39	0.815	72.50
241113 q	S1	2	20.6	4	3.87	0.28	0.433	73.00
241113 r	S1	2	20.5	6	5.98	0.20	0.279	73.00
241113 s	S1	2	20.8	8	8.08	0.09	0.171	72.50
241113 t	S1	2	20.7	8	7.94	0.17	0.217	72.50
241113 u	S1	2	20.7	8	7.78	0.26	0.248	72.50
241113 v	S1	2	20.7	8	8.16	0.41	0.288	72.50
241113 w	S1	2	20.6	20	20.29	0.17	0.124	72.00
241118 a	S1	2	18.4	10	10.23	0.26	0.230	71.50
241118 b	S1	2	18.7	12	12.35	0.29	0.216	71.00
241118 c	S1	2	18.8	14	14.16	0.27	0.199	71.50
241118 d	S1	2	18.9	16	16.45	0.46	0.232	71.50
241118 e	S1	2	18.9	16	16.63	0.36	0.203	71.00
241118 f	S1	2	18.9	16	15.92	0.23	0.171	71.00
241118 g	S1	2	18.9	16	16.15	0.08	0.112	71.50
241118 h	S1	2	18.9	20	20.59	0.31	0.171	71.00
241118 i	S1	2	19.0	24	24.31	0.20	0.128	71.50
241118 j	S1	2	19.0	28	28.28	0.23	0.127	72.00
241118 k	S1	2	19.1	32	33.29	0.54	0.177	71.50
241118 l	S1	2	19.1	32	33.14	0.37	0.148	71.50
241118 m	S1	2	19.2	32	32.91	0.29	0.131	71.50
241118 n	S1	2	19.2	32	32.25	0.15	0.101	72.00
241118 o	S1	2	19.4	32	32.72	0.06	0.067	71.00
241118 p	S1	2	19.4	40	40.34	0.18	0.098	72.00
241118 q	S1	2	19.8	50	51.08	0.15	0.081	71.50
241118 r	S1	2	19.7	75	78.71	0.23	0.084	71.00
241118 s	S1	2	19.7	100	103.38	0.23	0.076	72.00
241118 t	S1	2	19.8	150	154.13	0.22	0.066	71.50
241118 aa	S1	1	19.7	2	2.18	0.27	0.340	70.00
241118 ab	S1	1	20.0	4	4.05	0.23	0.197	71.00
241118 ad	S1	1	20.1	8	7.71	0.31	0.141	72.00
241118 ae	S1	1	20.0	10	9.86	0.27	0.113	71.00
241118 af	S1	1	20.0	12	12.32	0.14	0.081	72.00
241118 ag	S1	1	19.9	14	13.88	0.25	0.089	72.00
241118 ai	S1	1	20.0	20	20.05	0.11	0.053	73.00
241118 aj	S1	1	19.8	24	24.03	0.29	0.072	72.00

241118	ak	S1	1	20.1	28	29.16	0.37	0.073	70.00
241118	al	S1	1	20.1	32	32.54	0.26	0.061	72.00
241118	am	S1	1	20.1	40	41.06	0.24	0.052	72.00
241118	an	S1	1	20.1	50	50.27	0.22	0.045	72.00
241022	h	S2	2	22.4	2	2.08	0.37	1.434	81.00
241022	i	S2	2	22.4	2	2.08	0.30	1.310	80.00
241022	j	S2	2	22.5	2	2.03	0.23	1.209	80.50
241022	k	S2	2	22.2	2	2.03	0.15	1.031	80.00
241022	m	S2	2	22.4	8	8.35	0.39	0.801	80.50
241022	n	S2	2	22.5	8	8.23	0.25	0.664	73.50
241022	o	S2	2	22.5	8	8.19	0.13	0.481	73.50
241022	p	S2	2	22.6	8	8.20	0.14	0.498	73.50
241022	q	S2	2	22.5	8	8.40	0.40	0.805	73.50
241022	s	S2	2	22.1	32	33.59	0.47	0.537	73.50
241022	t	S2	2	22.4	32	33.50	0.33	0.441	73.50
241022	u	S2	2	22.5	32	33.14	0.20	0.342	71.50
241022	v	S2	2	22.5	32	32.92	0.08	0.196	71.50
241111	a	S2	2	20.6	16	16.41	0.16	0.453	70.50
241111	b	S2	2	21.0	16	16.47	0.16	0.426	70.50
241111	c	S2	2	21.3	16	16.56	0.23	0.513	70.50
241111	d	S2	2	21.3	16	16.62	0.30	0.560	70.50
241111	e	S2	2	21.4	16	16.72	0.45	0.685	70.50
241111	f	S2	2	21.4	16	16.72	0.45	0.689	70.50
241111	g	S2	2	21.7	4	4.14	0.21	0.814	70.50
241111	h	S2	2	21.8	6	6.26	0.29	0.775	70.50
241111	i	S2	2	21.7	10	10.46	0.29	0.646	70.50
241111	j	S2	2	21.7	12	12.54	0.31	0.615	70.50
241111	k	S2	2	21.8	14	14.65	0.30	0.566	70.50
241111	l	S2	2	21.9	20	20.85	0.33	0.531	70.50
241111	m	S2	2	21.8	24	25.09	0.40	0.535	70.50
241111	n	S2	2	22.2	28	29.13	0.28	0.401	70.50
241111	o	S2	2	21.9	40	41.52	0.20	0.288	70.50
241111	p	S2	2	22.4	50	51.97	0.24	0.296	70.50
241111	q	S2	2	22.3	60	62.22	0.38	0.345	70.50
241111	r	S2	2	22.3	70	73.04	0.31	0.294	70.50
241111	s	S2	2	22.4	100	105.17	0.40	0.290	70.50
241111	t	S2	2	22.5	150	156.77	0.31	0.207	70.50
250722	a	S2	2	20.2	4	4.11	0.28	1.000	100.00
250722	b	S2	2	20.1	4	4.20	0.37	1.092	99.50
250722	c	S2	2	20.8	4	4.12	0.33	1.044	100.00
250722	d	S2	2	21.3	4	4.14	0.35	1.041	100.50
250722	e	S2	2	21.4	4	4.07	0.20	0.832	100.00
250722	f	S2	2	21.5	50	51.02	0.38	0.394	100.00
250722	g	S2	2	22.0	50	51.06	0.36	0.385	100.00
250722	h	S2	2	22.7	50	51.13	0.26	0.318	100.00
250722	i	S2	2	22.6	50	51.29	0.28	0.324	100.00
250722	j	S2	2	22.8	50	50.96	0.25	0.310	100.50
250722	l	S2	2	21.4	90	93.26	0.30	0.260	100.00
250722	m	S2	2	21.6	100	103.29	0.37	0.285	100.00
250722	n	S2	2	21.9	100	103.37	0.39	0.293	100.00

250722	o	S2	2	22.1	100	102.96	0.35	0.268	100.00
250722	p	S2	2	22.5	100	104.05	0.22	0.212	99.50
250722	q	S2	2	22.4	100	103.30	0.24	0.215	100.00
250722	r	S2	2	23.1	110	112.41	0.28	0.223	100.00
250722	s	S2	2	23.1	120	122.77	0.29	0.217	100.50
250722	t	S2	2	23.0	130	130.96	0.34	0.231	100.50
250722	u	S2	2	23.0	140	141.27	0.28	0.202	100.50
250722	v	S2	2	23.5	150	154.90	0.42	0.247	100.00
250722	w	S2	2	23.7	150	155.25	0.42	0.250	100.50
250722	x	S2	2	23.5	150	152.79	0.34	0.215	100.00
250722	y	S2	2	23.4	150	155.05	0.34	0.218	100.00
250722	z	S2	2	23.7	150	153.27	0.30	0.202	100.00
250723	a	S2	1	21.3	2	1.93	0.31	0.713	109.00
250723	b	S2	1	21.4	2	2.07	0.45	0.821	168.00
250723	c	S2	1	21.4	2	2.02	0.28	0.679	167.00
250723	d	S2	1	21.5	4	4.15	0.36	0.529	167.00
250723	e	S2	1	21.6	4	4.10	0.40	0.565	169.00
250723	f	S2	1	21.6	4	3.95	0.22	0.441	168.00
250723	g	S2	1	21.9	6	5.90	0.22	0.383	169.00
250723	h	S2	1	21.8	8	8.11	0.18	0.283	168.00
250723	i	S2	1	21.8	8	8.33	0.38	0.402	167.00
250723	j	S2	1	21.8	8	8.13	0.30	0.356	169.00
250723	k	S2	1	21.9	12	12.07	0.25	0.282	118.00
250723	l	S2	1	21.9	16	16.59	0.40	0.314	117.00
250723	m	S2	1	22.0	16	16.74	0.43	0.316	116.00
250723	n	S2	1	21.9	16	16.51	0.47	0.335	118.00
250723	o	S2	1	22.0	24	24.88	0.32	0.235	117.00
250723	p	S2	1	22.0	32	32.39	0.26	0.187	118.00
250723	q	S2	1	22.1	24	24.52	0.45	0.279	119.00
250723	r	S2	1	22.2	32	32.80	0.35	0.219	117.00
250723	s	S2	1	21.5	40	40.63	0.39	0.214	118.00
250723	t	S2	1	20.8	50	51.14	0.21	0.152	117.00
250723	u	S2	1	21.8	50	50.92	0.38	0.185	117.00
250723	v	S2	1	22.0	50	50.90	0.24	0.151	117.00
250723	w	S2	1	21.9	100	103.66	0.29	0.121	117.00
250723	x	S2	1	22.2	100	104.45	0.43	0.150	116.00
250723	y	S2	1	22.3	100	104.09	0.28	0.122	116.00
241112	b	S3	2	21.1	2	2.07	0.09	2.797	81.00
241112	c	S3	2	21.2	2	2.06	0.18	3.699	81.50
241112	d	S3	2	21.3	2	2.08	0.43	5.590	81.50
241112	f	S3	2	21.8	4	4.10	0.28	3.241	81.50
241112	g	S3	2	22.4	6	6.16	0.36	3.023	81.50
241112	h	S3	2	22.7	8	8.20	0.33	2.789	81.50
241112	l	S3	2	22.6	10	10.20	0.33	2.312	82.00
241112	o	S3	2	22.9	16	16.43	0.07	0.997	82.00
241112	q	S3	2	21.3	16	16.40	0.26	1.885	81.50
241113	a	S3	2	19.6	16	16.33	0.11	1.369	70.50
241113	b	S3	2	20.0	16	16.27	0.58	3.454	70.50
241113	c	S3	2	20.1	16	16.74	0.19	1.650	23.50
241113	d	S3	2	20.3	10	10.19	0.26	2.461	70.50

241113	e	S3	2	20.4	13	13.27	0.25	2.108	70.50
241113	f	S3	2	20.6	20	20.47	0.18	1.449	70.50
241113	h	S3	2	20.6	28	28.76	0.27	1.490	70.50
241118	ba	S3	1	19.4	2	2.05	0.34	2.726	71.00
241118	bb	S3	1	19.5	4	4.02	0.28	1.804	72.00
241118	bc	S3	1	19.6	6	6.08	0.35	1.754	72.00
241118	bd	S3	1	19.3	8	7.93	0.33	1.480	73.00
241118	be	S3	1	19.6	10	10.22	0.34	1.355	71.00
241118	bf	S3	1	19.9	12	12.08	0.43	1.538	72.00
241118	bg	S3	1	19.9	16	16.09	0.42	1.279	72.00
241118	bh	S3	1	19.6	20	20.16	0.36	1.051	72.00
241118	bi	S3	1	19.9	24	24.55	0.45	1.136	71.00
241118	bj	S3	1	19.9	28	28.85	0.34	0.830	71.00
241118	bk	S3	1	19.8	32	32.58	0.22	0.640	72.00
250820	a	S4	2	22.5	2	2.05	0.31	2.569	115.50
250820	b	S4	2	22.5	2	2.07	0.43	3.023	115.00
250820	c	S4	2	22.7	2	2.05	0.27	2.470	115.00
250820	d	S4	2	22.7	4	4.05	0.34	2.067	115.00
250820	e	S4	2	23.2	4	4.11	0.32	2.023	100.00
250820	f	S4	2	21.2	4	4.13	0.36	2.132	100.00
250820	g	S4	2	23.0	8	8.20	0.41	1.756	100.00
250820	h	S4	2	21.1	8	8.22	0.31	1.519	100.00
250820	i	S4	2	22.5	8	8.21	0.17	1.145	100.00
250820	j	S4	2	22.7	16	16.38	0.44	1.370	100.00
250820	k	S4	2	23.3	16	16.43	0.25	0.992	100.00
250820	l	S4	2	23.3	16	16.44	0.38	1.242	100.00
250822	a	S4	2	21.8	32	32.86	0.33	0.860	103.00
250822	b	S4	2	21.9	32	32.60	0.45	1.058	103.50
250822	c	S4	2	22.0	32	33.07	0.32	0.844	103.00
250822	d	S4	2	22.2	40	40.87	0.23	0.640	103.50
250822	e	S4	2	22.4	50	50.89	0.31	0.679	104.00
250822	f	S4	2	22.5	50	51.18	0.31	0.678	103.50
250822	g	S4	2	22.5	50	51.09	0.37	0.746	103.50
250822	r	S4	2	21.9	6	6.09	0.31	1.711	107.00
250822	s	S4	2	21.9	12	12.19	0.35	1.369	107.00
250822	t	S4	2	22.4	20	20.33	0.30	1.017	107.00
250822	u	S4	2	22.3	24	24.54	0.25	0.829	107.00

SM3: Automated image analysis

Here, we present a comparison of automated and manual measurements of filament diameters, to verify the reliability of the automated measurements. Automated measurements of filament mid-plane diameters were made in MATLAB based on brightness thresholds to distinguish fluid and background, as stated in the main text. Manual measurements of filament diameters were made in Fiji ImageJ (Schindelin et al., 2012).

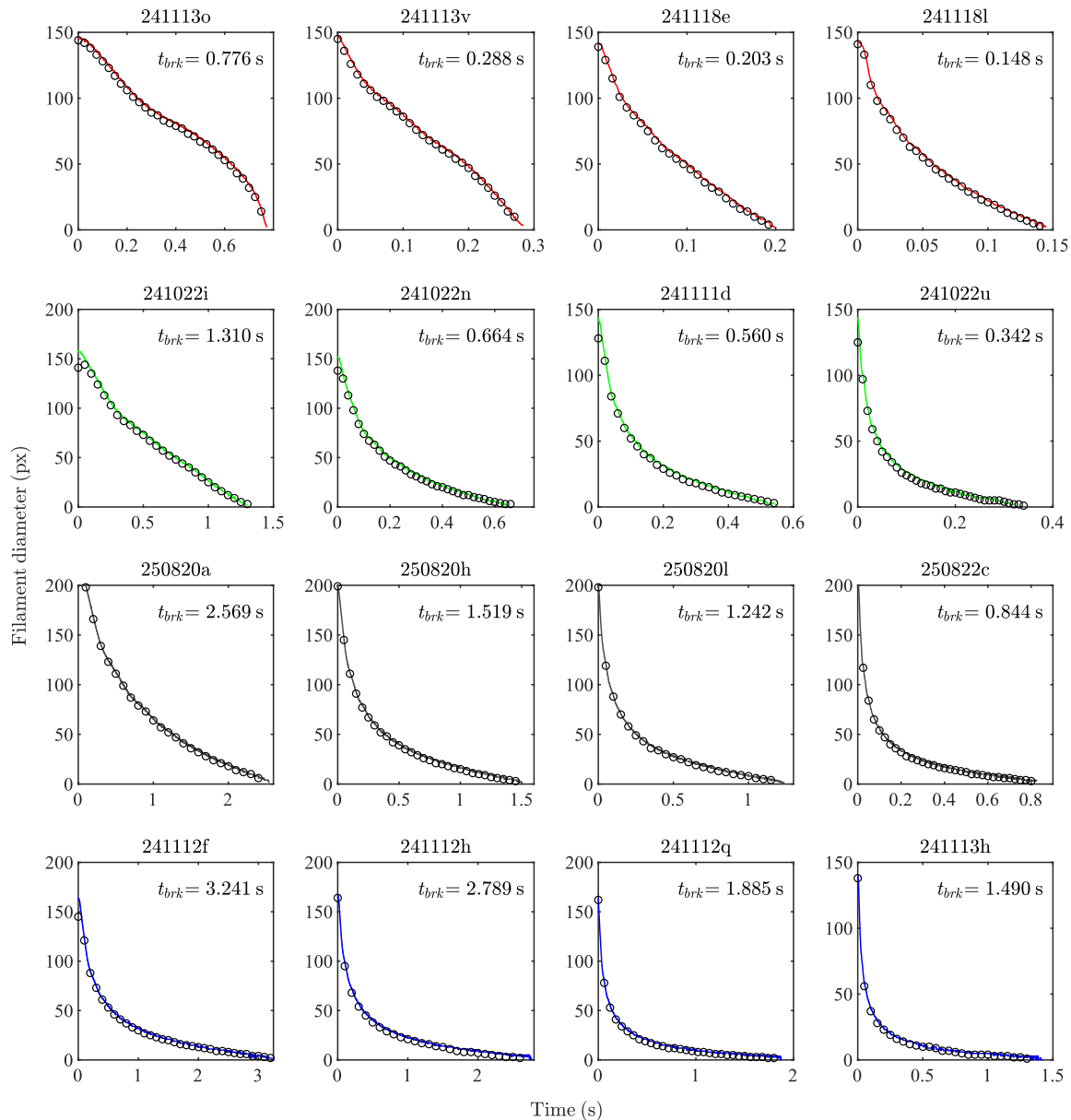


Figure A4. Comparison of automated measurements (solid lines: red = S1, green = S2, blue = S3, black = S4) and manual measurements (black circles). Images are labelled with the titles of corresponding video files.

SM4: Plate movement

The movement of the plates was generally smooth and constant, with only slight variations in speed throughout the stretching process. When plate separation is plotted against time, the trend can be described as linear (Figure A5). There is only minimal acceleration at the beginning of an experiment, which has a minor impact on thread thinning dynamics in experiments with a very short duration. For the vast majority of experiments, however, the plate separation velocity is essentially constant.

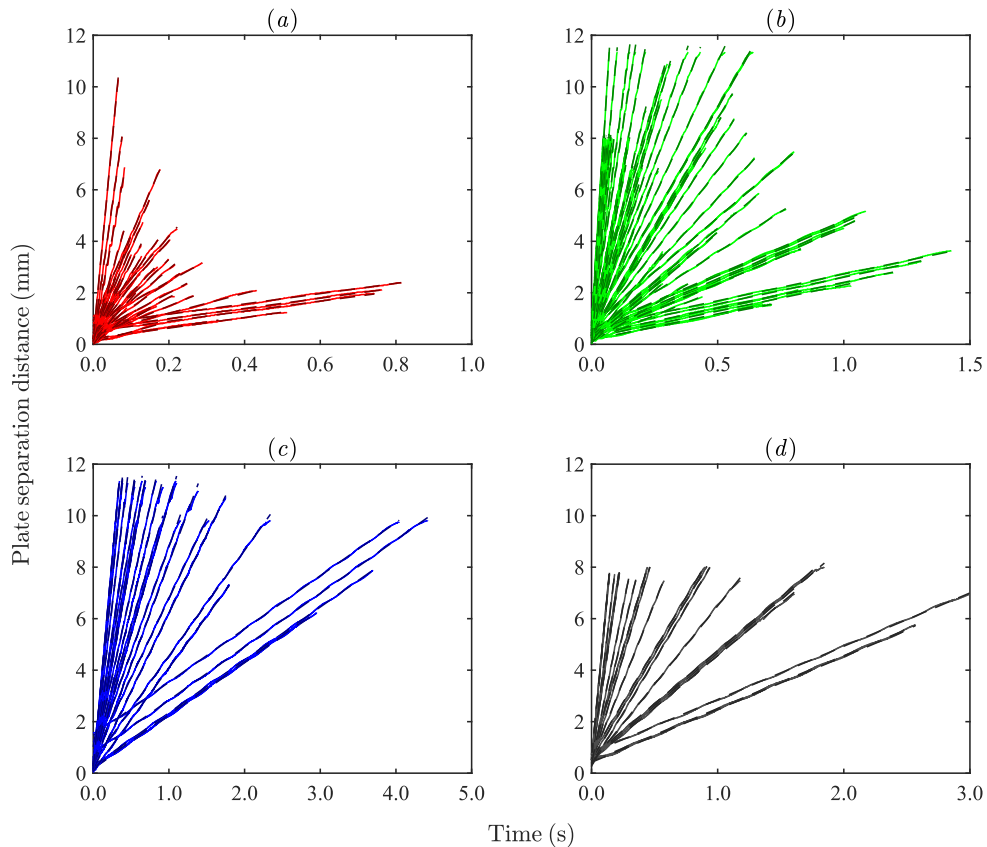


Figure A5. Tracking the positions of the plates over time for fluids: **a)** S1, **b)** S2, **c)** S3 and **d)** S4. Solid colour lines show plate positions, and dashed black lines show linear best-fits for calculating the average velocity. The maximum plate separation recorded depends on the camera field of view for a particular experiment.

In Figure A5, plate positions are tracked using semi-automated image analysis in MATLAB, similar to the measurement of the mid-filament diameter (Section 3.3). Plate edges are located based on a brightness threshold, which is manually specified to account for brightness differences between images. The initial plate separation depends on the volume of fluid used. The maximum recorded plate separation in this footage depends on the camera field of view for a particular experiment, and does not necessarily represent the final plate separation attained in that experiment. For the higher viscosity fluids S2, S3, and S4, breakup times were so long that the plates moved out of the field of view.

We calculate lines of best-fit through the data to find the average plate velocity for an experimental run. We find that the target plate velocity set on the TriMaster closely matches the plate velocity calculated from tracked plate positions (Figure A6).

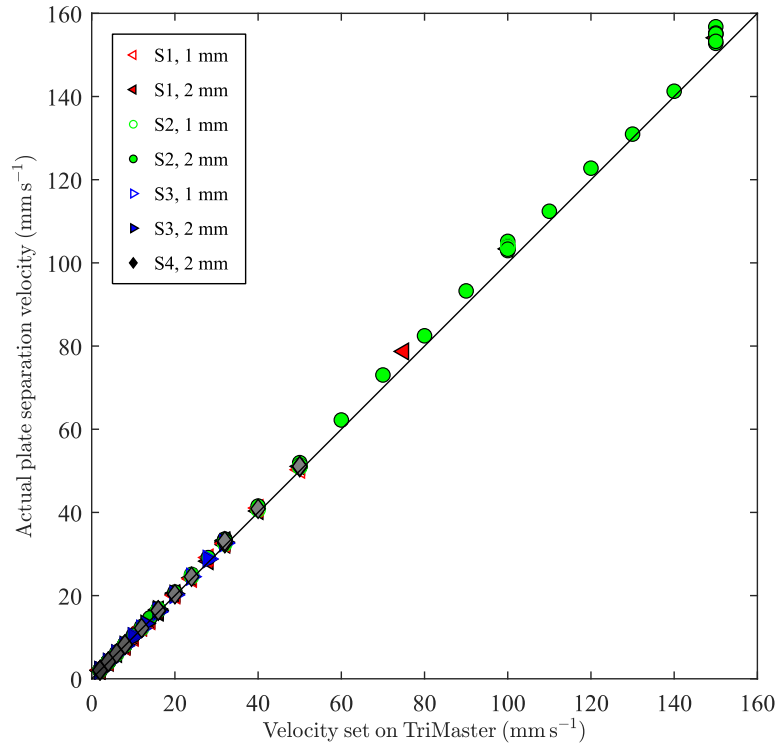


Figure A6. Comparison of target velocities set on the TriMaster extensional rheometer software, and the actual velocities attained by the plates, as measured from high-speed footage. Accuracy slightly decreases with increasing target velocity. Fluid type (S1-4) and plate diameter are distinguished by marker shape and colour.

References

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